Maximus Johnston

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Highly motivated Data Scientist with a proven record of delivering impactful analytics and machine learning solutions at scale across government, retail, and technology sectors. Skilled at uncovering insights in complex, large-scale datasets, shaping strategy, and securing stakeholder buy-in for transformative initiatives. Currently undertaking a Master's in Data Science at the University of Amsterdam to advance technical expertise and apply cutting-edge methods to real-world challenges.

Work Experience

Data Scientist (April 2023 – August 2025) - ***Department for Work and Pensions (DWP)***

Transforming Audit and Operational Efficiency (*SQL, Python, Databricks, git*)

- Scoped a high-value audit improvement project for health assessments, identifying opportunities for substantial savings for taxpayers and uncovering a reusable BAU process with a potential 6 figure saving
- Collaborated across audit, operational, and policy teams to secure senior approval for implementation, influencing national process changes.

Forecasting & Predictive Modelling (*PySpark, scikit-learn, Pandas, git*)

- Built and deployed forecasting models with >90% accuracy in predicting health service uptake, informing strategic workforce planning for thousands of staff and optimising multi-million-pound resource allocation

Advanced Analytics & Insight Discovery *(scikit-learn, NLP with Python, Pandas, git)*

- Discovered hidden claimant health-benefit patterns using K-means clustering and unstructured text analysis, enabling targeted interventions that improved service delivery for tens of thousands of claimants.

Stakeholder Engagement & Communication *(PowerBI, Looker, Matplotlib, git)*

- Partnered with operational and software teams to translate business needs into actionable data products, increasing adoption of data-driven decision-making across multiple programmes.
- Designed clear, accessible visualisations that accelerated programme-level decisions, reducing analysis-to-action time from weeks to days.

Data Science Intern (May - July 2021) - ***Data Clarity Limited, for a leading global British Fashion company***

Insight Generation from Large-Scale Retail Data *(Python, Pandas, Matplotlib, SQL, NetworkX, git)*

- Analysed 3M+ transactions to generate actionable insights, improving product recommendations and increasing targeted marketing effectiveness.
- Delivered market basket analysis visualised with a networkx diagram, revealing high-value product associations that informed retail strategy.

Predictive Modelling for Customer Behaviour *(scikit-learn, Pandas, Xgboost, SQL, git)*

- Developed predictive models to estimate next purchase dates with high accuracy, driving measurable sales uplift through optimised advertising and product placement

Undergraduate Software Engineer (September 2020- May 2021) - ***XACTPCB LTD, world leader in PCB registration solutions***

Feature Development & Product Improvement *(C#, XAML, WPF, git)*

- Delivered customer-facing features at scale within an Agile team, using C#, XAML, WPF, and MVVM/MVC design principles.
- Improved a large-scale PCB registration application used in over 150 factories worldwide through debugging, optimisation, and UX enhancements.
- Built interactive 3D visualisation tools for panel cross-sections, improving defect detection accuracy and manufacturing quality control.
- Implemented automated testing processes**,** enabling background execution and saving a full day of manual QA effort in every two-week sprint, boosting release reliability and freeing developer capacity for feature work.

Data Science Research Intern (July 2020- August 2020) - ***ENGIE*** *(Python data stack)*
Software and solutions architecture Intern (June - August 2019) - ***Excel Redstone/Google*** *(Java)*

Skills

Python, Python Data Stack, PySpark, SQL, Git, C#, Looker, Power BI, Java, Databricks, scikit-learn, XGBoost, Matplotlib, NetworkX

Education

University Van Amsterdam (August 2025 – Present)

- Master's in Data Science – advancing expertise in statistical modelling, AI, and large-scale data engineering.

Northumbria University (2023 - 2025)

- Postgraduate Diploma in Digital and Technology Solutions (Data Analytics pathway) – master's-level apprenticeship combining advanced data science with the creation and communication of innovative technological strategies and business proposals to senior stakeholders

Newcastle University (2018 - 2022)

- First-class degree in Computer Science with industrial placement (BSc). Including a first-class dissertation on the application of federated learning models created using TensorFlow
- I led a team which delivered a first-class application project to internal stakeholders

St Mary's Catholic High School, Chesterfield (2011 - 2018)

- A Levels: Mathematics, Chemistry, Physics

Achievements and extra-curricular

Grade 8 piano
Newcastle university men's Lacrosse team secretary
Ireland Tour - Cycled and camped the entire coast of Ireland travelling six weeks unsupported